

Namespace ASE_Assignment

Classes

[AppCanvas](#)

Class that implements the ICanvas interface and provides the functionality to draw on a canvas.

[AppCommandFactory](#)

Factory class for creating commands for the application.

[AppTriangle](#)

A command to draw a triangle with specified width and height.

[BOOSEInterpreter](#)

Main form for the BOOSE interpreter.

Interfaces

[ICanvas](#)

Interface for the application canvas.

Class AppCanvas

Namespace: [ASE Assignment](#)

Assembly: ASE Assignment.dll

Class that implements the ICanvas interface and provides the functionality to draw on a canvas.

```
public class AppCanvas : IAppCanvas, ICanvas
```

Inheritance

[object](#) ← AppCanvas

Implements

[IAppCanvas](#), ICanvas

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Constructors

AppCanvas()

Initializes a new instance of the [AppCanvas](#) class.

```
public AppCanvas()
```

Properties

PenColour

Gets or sets the color of the pen.

```
public object PenColour { get; set; }
```

Property Value

[object](#)

Xpos

Gets or sets the X position of the pen.

```
public int Xpos { get; set; }
```

Property Value

[int](#)

Ypos

Gets or sets the Y position of the pen.

```
public int Ypos { get; set; }
```

Property Value

[int](#)

Methods

Circle(int, bool)

Draws a circle on the canvas.

```
public void Circle(int radius, bool filled)
```

Parameters

radius [int](#)

The radius of the circle.

filled [bool](#)

If is set to true, the cricle will be filled.

Exceptions

[ArgumentOutOfRangeException](#)↗

Exception thrown when radius is equal to 0 or less.

Clear()

Clears the canvas to orginial state.

```
public void Clear()
```

DrawTo(int, int)

Draws a line from the current position to the specified position.

```
public void DrawTo(int x, int y)
```

Parameters

x [int](#)↗

The X coordinate to draw line to.

y [int](#)↗

The Y coordinate to draw line to.

Exceptions

[ArgumentOutOfRangeException](#)↗

Exception thrown when the coordinates given is out of the canvas size.

MoveTo(int, int)

Moves the pen to specified position without drawing on canvas.

```
public void MoveTo(int x, int y)
```

Parameters

x [int](#)

The X coordinate to move the pen to.

y [int](#)

The Y coordinate to move the pen to.

Exceptions

[ArgumentOutOfRangeException](#)

Exception thrown when coordinates given is out of the canvas size

Rect(int, int, bool)

Draw a rectangle on the canvas.

```
public void Rect(int width, int height, bool filled)
```

Parameters

width [int](#)

The width of the rectangle.

height [int](#)

The height of the rectangle.

filled [bool](#)

If is set to true, rectangle will be filled.

Exceptions

[ArgumentOutOfRangeException](#)

Exception thrown when width or height is set to 0 or less.

Reset()

Resets the pen position to original coordinates 0,0.

```
public void Reset()
```

Set(int, int)

Sets the size of the canvas.

```
public void Set(int xsize, int ysize)
```

Parameters

xsize [int](#)

The width of the canvas.

ysize [int](#)

The height of the canvas.

SetColour(int, int, int)

Sets the color of the pen.

```
public void SetColour(int red, int green, int blue)
```

Parameters

red [int](#)

Value of red component of the color

green [int](#)

Value of green component of the color

blue [int](#)

Value of blue component of the color

Exceptions

[ArgumentOutOfRangeException](#)

Exception thrown when any of the component values are set above 255.

Tri(int, int)

Draws a triangle on the canvas

```
public void Tri(int width, int height)
```

Parameters

width [int](#)

The width of the triangle.

height [int](#)

The height of the triangle.

Exceptions

[ArgumentOutOfRangeException](#)

Exception thrown when width or height is set to 0 or lower.

WriteText(string)

Writes the text on the canvas.

```
public void WriteText(string text)
```

Parameters

text [string](#) 

The text to write.

Exceptions

[NotImplementedException](#) 

Exception thrown because the method is not implemented.

getBitmap()

Gets the bitmap of the canvas.

```
public object getBitmap()
```

Returns

[object](#) 

The bitmap of the canvas.

Class AppCommandFactory

Namespace: [ASE Assignment](#)

Assembly: ASE Assignment.dll

Factory class for creating commands for the application.

```
public class AppCommandFactory : CommandFactory, ICommandFactory
```

Inheritance

[object](#)  ← CommandFactory ← AppCommandFactory

Implements

ICommandFactory

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

AppCommandFactory(IAppCanvas)

Initializes a new instance of the [AppCommandFactory](#) class.

```
public AppCommandFactory(IAppCanvas canvas)
```

Parameters

canvas [IAppCanvas](#)

The canvas to be used by the commands.

Methods

MakeCommand(string)

Creates a command based on the specified command type.

```
public override ICommand MakeCommand(string commandType)
```

Parameters

commandType [string](#) 

The type of command to be created.

Returns

ICommand

A instance of BOOSE.ICommandcorresponding to the specified command type.

Class AppTriangle

Namespace: [ASE Assignment](#)

Assembly: ASE Assignment.dll

A command to draw a triangle with specified width and height.

```
public class AppTriangle : ICommand
```

Inheritance

[object](#)  ← AppTriangle

Implements

ICommand

Inherited Members

[object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) , [object.GetHashCode\(\)](#) , [object.GetType\(\)](#) ,
[object.MemberwiseClone\(\)](#) , [object.ReferenceEquals\(object, object\)](#) , [object.ToString\(\)](#) 

Constructors

AppTriangle(IAppCanvas, int, int)

Initializes a new instance of the [AppTriangle](#) class.

```
public AppTriangle(IAppCanvas canvas, int width, int height)
```

Parameters

canvas [IAppCanvas](#)

The canvas to draw the triangle on.

width [int](#) 

The width of the triangle.

height [int](#) 

The height of the triangle.

Fields

height

```
public int height
```

Field Value

[int](#)

width

```
public int width
```

Field Value

[int](#)

Methods

CheckParameters(string[])

Check the parameters for the triangle command.

```
public void CheckParameters(string[] Parameters)
```

Parameters

Parameters [string](#)[]

The parameters that have to be checked.

Exceptions

[NotImplementedException](#)

Exception is thrown because this method is not implemented.

Compile()

Compiles the triangle command

```
public void Compile()
```

Execute()

Executes the triangle command.

```
public void Execute()
```

Set(StoredProgram, string)

Sets the parameters for the triangle command.

```
public void Set(StoredProgram Program, string Parameters)
```

Parameters

Program `StoredProgram`

The stored program

Parameters `string` [↗](#)

Exceptions

[Exception](#) [↗](#)

Exception thrown when there is less than 2 parameters given.

[ArgumentOutOfRangeException](#) [↗](#)

Exception thrown when the width or height of the triangle is equal to 0 or less.

Class BOOSEInterpreter

Namespace: [ASE Assignment](#)

Assembly: ASE Assignment.dll

Main form for the BOOSE interpreter.

```
public class BOOSEInterpreter : Form, IDropTarget, ISynchronizeInvoke, IWin32Window,
IBindableComponent, IComponent, IDisposable, IContainerControl
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← [ScrollableControl](#) ← [ContainerControl](#) ← [Form](#) ← BOOSEInterpreter

Implements

[IDropTarget](#), [ISynchronizeInvoke](#), [IWin32Window](#), [IBindableComponent](#), [IComponent](#), [IDisposable](#), [IContainerControl](#)

Inherited Members

[Form.SetVisibleCore\(bool\)](#), [Form.Activate\(\)](#), [Form.ActivateMdiChild\(Form\)](#), [Form.AddOwnedForm\(Form\)](#), [Form.AdjustFormScrollbars\(bool\)](#), [Form.Close\(\)](#), [Form.CreateAccessibilityInstance\(\)](#), [Form.CreateControlsInstance\(\)](#), [Form.CreateHandle\(\)](#), [Form.DefWndProc\(ref Message\)](#), [Form.ProcessMnemonic\(char\)](#), [Form.CenterToParent\(\)](#), [Form.CenterToScreen\(\)](#), [Form.LayoutMdi\(MdiLayout\)](#), [Form.OnActivated\(EventArgs\)](#), [Form.OnBackgroundImageChanged\(EventArgs\)](#), [Form.OnBackgroundImageLayoutChanged\(EventArgs\)](#), [Form.OnClosing\(CancelEventArgs\)](#), [Form.OnClosed\(EventArgs\)](#), [Form.OnFormClosing\(FormClosingEventArgs\)](#), [Form.OnFormClosed\(FormClosedEventArgs\)](#), [Form.OnCreateControl\(\)](#), [Form.OnDeactivate\(EventArgs\)](#), [Form.OnEnabledChanged\(EventArgs\)](#), [Form.OnEnter\(EventArgs\)](#), [Form.OnFontChanged\(EventArgs\)](#), [Form.OnGotFocus\(EventArgs\)](#), [Form.OnHandleCreated\(EventArgs\)](#), [Form.OnHandleDestroyed\(EventArgs\)](#), [Form.OnHelpButtonClicked\(CancelEventArgs\)](#), [Form.OnLayout\(LayoutEventArgs\)](#), [Form.OnLoad\(EventArgs\)](#), [Form.OnMaximizedBoundsChanged\(EventArgs\)](#), [Form.OnMaximumSizeChanged\(EventArgs\)](#), [Form.OnMinimumSizeChanged\(EventArgs\)](#), [Form.OnInputLanguageChanged\(InputLanguageChangedEventArgs\)](#), [Form.OnInputLanguageChanging\(InputLanguageChangingEventArgs\)](#), [Form.OnVisibleChanged\(EventArgs\)](#), [Form.OnMdiChildActivate\(EventArgs\)](#), [Form.OnMenuStart\(EventArgs\)](#), [Form.OnMenuComplete\(EventArgs\)](#), [Form.OnPaint\(PaintEventArgs\)](#), [Form.OnResize\(EventArgs\)](#), [Form.OnDpiChanged\(DpiChangedEventArgs\)](#), [Form.OnGetDpiScaledSize\(int, int, ref Size\)](#),

[Form.OnRightToLeftLayoutChanged\(EventArgs\)](#), [Form.OnShown\(EventArgs\)](#),
[Form.OnTextChanged\(EventArgs\)](#), [Form.ProcessCmdKey\(ref Message, Keys\)](#),
[Form.ProcessDialogKey\(Keys\)](#), [Form.ProcessDialogChar\(char\)](#),
[Form.ProcessKeyPreview\(ref Message\)](#), [Form.ProcessTabKey\(bool\)](#),
[Form.RemoveOwnedForm\(Form\)](#), [Form.Select\(bool, bool\)](#),
[Form.ScaleMinMaxSize\(float, float, bool\)](#),
[Form.GetScaledBounds\(Rectangle, SizeF, BoundsSpecified\)](#),
[Form.ScaleControl\(SizeF, BoundsSpecified\)](#), [Form.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#),
[Form.SetClientSizeCore\(int, int\)](#), [Form.SetDesktopBounds\(int, int, int, int\)](#),
[Form.SetDesktopLocation\(int, int\)](#), [Form.Show\(IWin32Window\)](#), [Form.ShowDialog\(\)](#),
[Form.ShowDialog\(IWin32Window\)](#), [Form.ToString\(\)](#), [Form.UpdateDefaultButton\(\)](#),
[Form.OnResizeBegin\(EventArgs\)](#), [Form.OnResizeEnd\(EventArgs\)](#),
[Form.OnStyleChanged\(EventArgs\)](#), [Form.ValidateChildren\(\)](#),
[Form.ValidateChildren\(ValidationConstraints\)](#), [Form.WndProc\(ref Message\)](#), [Form.AcceptButton](#),
[Form.ActiveForm](#), [Form.ActiveMdiChild](#), [Form.AllowTransparency](#), [Form.AutoScroll](#),
[Form.AutoSize](#), [Form.AutoSizeMode](#), [Form.AutoValidate](#), [Form.BackColor](#),
[Form.FormBorderStyle](#), [Form.CancelButton](#), [Form.ClientSize](#), [Form.ControlBox](#),
[Form.CreateParams](#), [Form.DefaultImeMode](#), [Form.DefaultSize](#), [Form.DesktopBounds](#),
[Form.DesktopLocation](#), [Form.DialogResult](#), [Form.HelpButton](#), [Form.Icon](#), [Form.IsMdiChild](#),
[Form.IsMdiContainer](#), [Form.IsRestrictedWindow](#), [Form.KeyPreview](#), [Form.Location](#),
[Form.MaximizedBounds](#), [Form.MaximumSize](#), [Form.MainMenuStrip](#), [Form.MinimumSize](#),
[Form.MaximizeBox](#), [Form.MdiChildren](#), [Form.MdiChildrenMinimizedAnchorBottom](#),
[Form.MdiParent](#), [Form.MinimizeBox](#), [Form.Modal](#), [Form.Opacity](#), [Form.OwnedForms](#),
[Form.Owner](#), [Form.RestoreBounds](#), [Form.RightToLeftLayout](#), [Form.ShowInTaskbar](#),
[Form.ShowIcon](#), [Form.ShowWithoutActivation](#), [Form.Size](#), [Form.SizeGripStyle](#),
[Form.StartPosition](#), [Form.Text](#), [Form.TopLevel](#), [Form.TopMost](#), [Form.TransparencyKey](#),
[Form.WindowState](#), [Form.AutoSizeChanged](#), [Form.AutoValidateChanged](#),
[Form.HelpButtonClicked](#), [Form.MaximizedBoundsChanged](#), [Form.MaximumSizeChanged](#),
[Form.MinimumSizeChanged](#), [Form.Activated](#), [Form.Deactivate](#), [Form.FormClosing](#),
[Form.FormClosed](#), [Form.Load](#), [Form.MdiChildActivate](#), [Form.MenuComplete](#),
[Form.MenuStart](#), [Form.InputLanguageChanged](#), [Form.InputLanguageChanging](#),
[Form.RightToLeftLayoutChanged](#), [Form.Shown](#), [Form.DpiChanged](#), [Form.ResizeBegin](#),
[Form.ResizeEnd](#), [ContainerControl.OnAutoValidateChanged\(EventArgs\)](#),
[ContainerControl.OnMove\(EventArgs\)](#), [ContainerControl.OnParentChanged\(EventArgs\)](#),
[ContainerControl.PerformAutoScale\(\)](#), [ContainerControl.RescaleConstantsForDpi\(int, int\)](#),
[ContainerControl.Validate\(\)](#), [ContainerControl.Validate\(bool\)](#),
[ContainerControl.AutoScaleDimensions](#), [ContainerControl.AutoScaleFactor](#),
[ContainerControl.AutoScaleMode](#), [ContainerControl.BindingContext](#),
[ContainerControl.CanEnableIme](#), [ContainerControl.ActiveControl](#),
[ContainerControl.CurrentAutoScaleDimensions](#), [ContainerControl.ParentForm](#),

[ScrollableControl.ScrollStateAutoScrolling](#) , [ScrollableControl.ScrollStateHScrollVisible](#) ,
[ScrollableControl.ScrollStateVScrollVisible](#) , [ScrollableControl.ScrollStateUserHasScrolled](#) ,
[ScrollableControl.ScrollStateFullDrag](#) , [ScrollableControl.GetScrollState\(int\)](#) ,
[ScrollableControl.OnMouseWheel\(MouseEventArgs\)](#) ,
[ScrollableControl.OnRightToLeftChanged\(EventArgs\)](#) ,
[ScrollableControl.OnPaintBackground\(PaintEventArgs\)](#) ,
[ScrollableControl.OnPaddingChanged\(EventArgs\)](#) , [ScrollableControl.SetDisplayRectLocation\(int, int\)](#) ,
[ScrollableControl.ScrollControlIntoView\(Control\)](#) , [ScrollableControl.ScrollToControl\(Control\)](#) ,
[ScrollableControl.OnScroll\(ScrollEventArgs\)](#) , [ScrollableControl.SetAutoScrollMargin\(int, int\)](#) ,
[ScrollableControl.SetScrollState\(int, bool\)](#) , [ScrollableControl.AutoScrollMargin](#) ,
[ScrollableControl.AutoScrollPosition](#) , [ScrollableControl.AutoScrollMinSize](#) ,
[ScrollableControl.DisplayRectangle](#) , [ScrollableControl.HScroll](#) , [ScrollableControl.HorizontalScroll](#) ,
[ScrollableControl.VScroll](#) , [ScrollableControl.VerticalScroll](#) , [ScrollableControl.Scroll](#) ,
[Control.GetAccessibilityObjectById\(int\)](#) , [Control.SetAutoSizeMode\(AutoSizeMode\)](#) ,
[Control.GetAutoSizeMode\(\)](#) , [Control.GetPreferredSize\(Size\)](#) ,
[Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#) ,
[Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#) , [Control.BeginInvoke\(Delegate\)](#) ,
[Control.BeginInvoke\(Action\)](#) , [Control.BeginInvoke\(Delegate, params object\[\]\)](#) ,
[Control.BringToFront\(\)](#) , [Control.Contains\(Control\)](#) , [Control.CreateGraphics\(\)](#) ,
[Control.CreateControl\(\)](#) , [Control.DestroyHandle\(\)](#) , [Control.DoDragDrop\(object, DragDropEffects\)](#) ,
[Control.DoDragDrop\(object, DragDropEffects, Bitmap, Point, bool\)](#) ,
[Control.DrawToBitmap\(Bitmap, Rectangle\)](#) , [Control.EndInvoke\(IAsyncResult\)](#) , [Control.FindForm\(\)](#) ,
[Control.GetTopLevel\(\)](#) , [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#) ,
[Control.RaiseMouseEvent\(object, MouseEventArgs\)](#) , [Control.Focus\(\)](#) ,
[Control.FromChildHandle\(nint\)](#) , [Control.FromHandle\(nint\)](#) ,
[Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#) , [Control.GetChildAtPoint\(Point\)](#) ,
[Control.GetContainerControl\(\)](#) , [Control.GetNextControl\(Control, bool\)](#) ,
[Control.GetStyle\(ControlStyles\)](#) , [Control.Hide\(\)](#) , [Control.InitLayout\(\)](#) , [Control.Invalidate\(Region\)](#) ,
[Control.Invalidate\(Region, bool\)](#) , [Control.Invalidate\(\)](#) , [Control.Invalidate\(bool\)](#) ,
[Control.Invalidate\(Rectangle\)](#) , [Control.Invalidate\(Rectangle, bool\)](#) , [Control.Invoke\(Action\)](#) ,
[Control.Invoke\(Delegate\)](#) , [Control.Invoke\(Delegate, params object\[\]\)](#) ,
[Control.Invoke<T>\(Func<T>\)](#) , [Control.InvokePaint\(Control, PaintEventArgs\)](#) ,
[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#) , [Control.IsKeyLocked\(Keys\)](#) ,
[Control.IsInputChar\(char\)](#) , [Control.IsInputKey\(Keys\)](#) , [Control.IsMnemonic\(char, string\)](#) ,
[Control.LogicalToDeviceUnits\(int\)](#) , [Control.LogicalToDeviceUnits\(Size\)](#) ,
[Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#) , [Control.NotifyInvalidate\(Rectangle\)](#) ,
[Control.InvokeOnClick\(Control, EventArgs\)](#) , [Control.OnAutoSizeChanged\(EventArgs\)](#) ,
[Control.OnBackColorChanged\(EventArgs\)](#) , [Control.OnBindingContextChanged\(EventArgs\)](#) ,
[Control.OnCausesValidationChanged\(EventArgs\)](#) , [Control.OnContextMenuStripChanged\(EventArgs\)](#) ,
[Control.OnCursorChanged\(EventArgs\)](#) , [Control.OnDataContextChanged\(EventArgs\)](#) ,

[Control.OnDockChanged\(EventArgs\)](#), [Control.OnForeColorChanged\(EventArgs\)](#),
[Control.OnNotifyMessage\(Message\)](#), [Control.OnParentBackColorChanged\(EventArgs\)](#),
[Control.OnParentBackgroundImageChanged\(EventArgs\)](#),
[Control.OnParentBindingContextChanged\(EventArgs\)](#), [Control.OnParentCursorChanged\(EventArgs\)](#),
[Control.OnParentDataContextChanged\(EventArgs\)](#), [Control.OnParentEnabledChanged\(EventArgs\)](#),
[Control.OnParentFontChanged\(EventArgs\)](#), [Control.OnParentForeColorChanged\(EventArgs\)](#),
[Control.OnParentRightToLeftChanged\(EventArgs\)](#), [Control.OnParentVisibleChanged\(EventArgs\)](#),
[Control.OnPrint\(PaintEventArgs\)](#), [Control.OnTabIndexChanged\(EventArgs\)](#),
[Control.OnTabStopChanged\(EventArgs\)](#), [Control.OnClick\(EventArgs\)](#),
[Control.OnClientSizeChanged\(EventArgs\)](#), [Control.OnControlAdded\(ControlEventArgs\)](#),
[Control.OnControlRemoved\(ControlEventArgs\)](#), [Control.OnLocationChanged\(EventArgs\)](#),
[Control.OnDoubleClick\(EventArgs\)](#), [Control.OnDragEnter\(DragEventArgs\)](#),
[Control.OnDragOver\(DragEventArgs\)](#), [Control.OnDragLeave\(EventArgs\)](#),
[Control.OnDragDrop\(DragEventArgs\)](#), [Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#),
[Control.InvokeGotFocus\(Control, EventArgs\)](#), [Control.OnHelpRequested\(HelpEventArgs\)](#),
[Control.OnInvalidated\(InvalidateEventArgs\)](#), [Control.OnKeyDown\(KeyEventArgs\)](#),
[Control.OnKeyPress\(KeyPressEventArgs\)](#), [Control.OnKeyUp\(KeyEventArgs\)](#),
[Control.OnLeave\(EventArgs\)](#), [Control.InvokeLostFocus\(Control, EventArgs\)](#),
[Control.OnLostFocus\(EventArgs\)](#), [Control.OnMarginChanged\(EventArgs\)](#),
[Control.OnMouseDoubleClick\(MouseEventArgs\)](#), [Control.OnMouseClick\(MouseEventArgs\)](#),
[Control.OnMouseCaptureChanged\(EventArgs\)](#), [Control.OnMouseDown\(MouseEventArgs\)](#),
[Control.OnMouseEnter\(EventArgs\)](#), [Control.OnMouseLeave\(EventArgs\)](#),
[Control.OnDpiChangedBeforeParent\(EventArgs\)](#), [Control.OnDpiChangedAfterParent\(EventArgs\)](#),
[Control.OnMouseHover\(EventArgs\)](#), [Control.OnMouseMove\(MouseEventArgs\)](#),
[Control.OnMouseUp\(MouseEventArgs\)](#),
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#),
[Control.OnRegionChanged\(EventArgs\)](#), [Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#),
[Control.OnSizeChanged\(EventArgs\)](#), [Control.OnChangeUICues\(UICuesEventArgs\)](#),
[Control.OnSystemColorsChanged\(EventArgs\)](#), [Control.OnValidating\(CancelEventArgs\)](#),
[Control.OnValidated\(EventArgs\)](#), [Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#),
[Control.PointToClient\(Point\)](#), [Control.PointToScreen\(Point\)](#),
[Control.PreProcessMessage\(ref Message\)](#), [Control.PreProcessControlMessage\(ref Message\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#), [Control.RaisePaintEvent\(object, PaintEventArgs\)](#),
[Control.RecreateHandle\(\)](#), [Control.RectangleToClient\(Rectangle\)](#),
[Control.RectangleToScreen\(Rectangle\)](#), [Control.ReflectMessage\(nint, ref Message\)](#),
[Control.Refresh\(\)](#), [Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.Select\(\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),

[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#), [Control.SetTopLevel\(bool\)](#),
[Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),
[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#), [Control.RtlTranslateContent\(ContentAlignment\)](#),
[Control.Show\(\)](#), [Control.SuspendLayout\(\)](#), [Control.Update\(\)](#), [Control.UpdateBounds\(\)](#),
[Control.UpdateBounds\(int, int, int, int\)](#), [Control.UpdateBounds\(int, int, int, int, int, int\)](#),
[Control.UpdateZOrder\(\)](#), [Control.UpdateStyles\(\)](#), [Control.OnImeModeChanged\(EventArgs\)](#),
[Control.AccessibilityObject](#), [Control.AccessibleDefaultActionDescription](#),
[Control.AccessibleDescription](#), [Control.AccessibleName](#), [Control.AccessibleRole](#),
[Control.AllowDrop](#), [Control.Anchor](#), [Control.AutoScrollOffset](#), [Control.LayoutEngine](#),
[Control.DataContext](#), [Control.BackgroundImage](#), [Control.BackgroundImageLayout](#),
[Control.Bottom](#), [Control.Bounds](#), [Control.CanFocus](#), [Control.CanRaiseEvents](#),
[Control.CanSelect](#), [Control.Capture](#), [Control.CausesValidation](#),
[Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#), [Control.CompanyName](#),
[Control.ContainsFocus](#), [Control.ContextMenuStrip](#), [Control.Controls](#), [Control.Created](#),
[Control.Cursor](#), [Control.DataBindings](#), [Control.DefaultBackColor](#), [Control.DefaultCursor](#),
[Control.DefaultFont](#), [Control.DefaultForeColor](#), [Control.DefaultMargin](#),
[Control.DefaultMaximumSize](#), [Control.DefaultMinimumSize](#), [Control.DefaultPadding](#),
[Control.DeviceDpi](#), [Control.IsDisposed](#), [Control.Disposing](#), [Control.Dock](#),
[Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#), [Control.Font](#),
[Control.FontHeight](#), [Control.ForeColor](#), [Control.Handle](#), [Control.HasChildren](#), [Control.Height](#),
[Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#),
[Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Margin](#),
[Control.ModifierKeys](#), [Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#),
[Control.Parent](#), [Control.ProductName](#), [Control.ProductVersion](#), [Control.RecreatingHandle](#),
[Control.Region](#), [Control.RenderRightToLeft](#), [Control.ResizeRedraw](#), [Control.Right](#),
[Control.RightToLeft](#), [Control.ScaleChildren](#), [Control.Site](#), [Control.TabIndex](#), [Control.TabStop](#),
[Control.Tag](#), [Control.Top](#), [Control.TopLevelControl](#), [Control.ShowKeyboardCues](#),
[Control.ShowFocusCues](#), [Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#),
[Control.PreferredSize](#), [Control.Padding](#), [Control.ImeMode](#), [Control.ImeModeBase](#),
[Control.PropagatingImeMode](#), [Control.BackColorChanged](#), [Control.BackgroundImageChanged](#),
[Control.BackgroundImageLayoutChanged](#), [Control.BindingContextChanged](#),
[Control.CausesValidationChanged](#), [Control.ClientSizeChanged](#),
[Control.ContextMenuStripChanged](#), [Control.CursorChanged](#), [Control.DockChanged](#),
[Control.EnabledChanged](#), [Control.FontChanged](#), [Control.ForeColorChanged](#),
[Control.LocationChanged](#), [Control.MarginChanged](#), [Control.RegionChanged](#),
[Control.RightToLeftChanged](#), [Control.SizeChanged](#), [Control.TabIndexChanged](#),
[Control.TabStopChanged](#), [Control.TextChanged](#), [Control.VisibleChanged](#), [Control.Click](#),

[Control.ControlAdded](#) , [Control.ControlRemoved](#) , [Control.DataContextChanged](#) ,
[Control.DragDrop](#) , [Control.DragEnter](#) , [Control.DragOver](#) , [Control.DragLeave](#) ,
[Control.GiveFeedback](#) , [Control.HandleCreated](#) , [Control.HandleDestroyed](#) ,
[Control.HelpRequested](#) , [Control.Invalidated](#) , [Control.PaddingChanged](#) , [Control.Paint](#) ,
[Control.QueryContinueDrag](#) , [Control.QueryAccessibilityHelp](#) , [Control.DoubleClick](#) ,
[Control.Enter](#) , [Control.GotFocus](#) , [Control.KeyDown](#) , [Control.KeyPress](#) , [Control.KeyUp](#) ,
[Control.Layout](#) , [Control.Leave](#) , [Control.LostFocus](#) , [Control.MouseClick](#) ,
[Control.MouseDoubleClick](#) , [Control.MouseCaptureChanged](#) , [Control.MouseDown](#) ,
[Control.MouseEnter](#) , [Control.MouseLeave](#) , [Control.DpiChangedBeforeParent](#) ,
[Control.DpiChangedAfterParent](#) , [Control.MouseHover](#) , [Control.MouseMove](#) , [Control.MouseUp](#) ,
[Control.MouseWheel](#) , [Control.Move](#) , [Control.PreviewKeyDown](#) , [Control.Resize](#) ,
[Control.ChangeUICues](#) , [Control.StyleChanged](#) , [Control.SystemColorsChanged](#) ,
[Control.Validating](#) , [Control.Validated](#) , [Control.ParentChanged](#) , [Control.ImeModeChanged](#) ,
[Component.Dispose\(\)](#) , [Component.GetService\(Type\)](#) , [Component.Container](#) ,
[Component.DesignMode](#) , [Component.Events](#) , [Component.Disposed](#) ,
[MarshalByRefObject.GetLifetimeService\(\)](#) , [MarshalByRefObject.InitializeLifetimeService\(\)](#) ,
[MarshalByRefObject.MemberwiseClone\(bool\)](#) , [object.Equals\(object\)](#) , [object.Equals\(object, object\)](#) ,
[object.GetHashCode\(\)](#) , [object.GetType\(\)](#) , [object.MemberwiseClone\(\)](#) ,
[object.ReferenceEquals\(object, object\)](#)

Constructors

BOOSEInterpreter()

```
public BOOSEInterpreter()
```

Methods

Dispose(bool)

Clean up any resources being used.

```
protected override void Dispose(bool disposing)
```

Parameters

disposing [bool](#)

true if managed resources should be disposed; otherwise, false.

Interface IAppCanvas

Namespace: [ASE Assignment](#)

Assembly: ASE Assignment.dll

Interface for the application canvas.

```
public interface IAppCanvas : ICanvas
```

Inherited Members

[ICanvas.Set\(int, int\)](#) , [ICanvas.SetColour\(int, int, int\)](#) , [ICanvas.MoveTo\(int, int\)](#) , [ICanvas.DrawTo\(int, int\)](#) , [ICanvas.Clear\(\)](#) , [ICanvas.Reset\(\)](#) , [ICanvas.Circle\(int, bool\)](#) , [ICanvas.Rect\(int, int, bool\)](#) , [ICanvas.WriteText\(string\)](#) , [ICanvas.getBitmap\(\)](#) , [ICanvas.Xpos](#) , [ICanvas.Ypos](#) , [ICanvas.PenColour](#)

Methods

Tri(int, int)

Draws a triangle on the canvas.

```
void Tri(int width, int height)
```

Parameters

width [int](#)

The width of the triangle.

height [int](#)

The height of the triangle.

Namespace AppCanvasTests

Classes

[CommandsTest](#)

Unit test for the [AppCanvas](#) class.

Class CommandsTest

Namespace: [AppCanvasTests](#)

Assembly: AppCanvasTests.dll

Unit test for the [AppCanvas](#) class.

```
[TestClass]
public class CommandsTest
```

Inheritance

[object](#) ← CommandsTest

Inherited Members

[object.Equals\(object\)](#), [object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#), [object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#), [object.ToString\(\)](#)

Methods

Circle_Test_ValidRadius()

Test for / Test for [Circle\(int, bool\)](#) method with valid radius.

```
[TestMethod]
public void Circle_Test_ValidRadius()
```

DrawTo_Test()

Test for the AppCanvas.DrawLineTo(int, int) method with valid coordinates.

```
[TestMethod]
public void DrawTo_Test()
```

DrawTo_Test_Negative_Coords()

Test for the `AppCanvas.DrawLineTo(int, int)` method with negative coordinates. Exception [ArgumentOutOfRangeException](#) expected to be thrown.

```
[TestMethod]
[ExpectedException(typeof(ArgumentOutOfRangeException))]
public void DrawTo_Test_Negative_Coords()
```

MoveTo_Test()

Test for the `MoveTo(int, int)` method with valid coordinates.

```
[TestMethod]
public void MoveTo_Test()
```

MoveTo_Test_Negative_Coords()

Test for the `MoveTo(int, int)` method with negative coordinates. Exception [ArgumentOutOfRangeException](#) expected to be thrown.

```
[TestMethod]
[ExpectedException(typeof(ArgumentOutOfRangeException))]
public void MoveTo_Test_Negative_Coords()
```

Namespace AppTriangleTests

Classes

[CommandTests](#)

Class CommandTests

Namespace: [AppTriangleTests](#)

Assembly: AppTriangleTests.dll

```
[TestClass]
public class CommandTests
```

Inheritance

[object](#)  ← CommandTests

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  ,
[object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Methods

Correct_Parameters_Check()

```
[TestMethod]
public void Correct_Parameters_Check()
```

Invalid_Parameters_Check()

```
[TestMethod]
[ExpectedException(typeof(Exception))]
public void Invalid_Parameters_Check()
```

Paramater_0_Check()

```
[TestMethod]
[ExpectedException(typeof(ArgumentOutOfRangeException))]
public void Paramater_0_Check()
```